Shahzeb Jadoon

AI / ML Engineer

linkedin.com/in/shahzebkjadoon
shahzeb.jadoon.sj@gmail.com
(585) 202-9088
Rochester, NY

SUMMARY

Edge-AI and ML engineer specializing in hardware-software co-design - deploying deep models on constrained and neuromorphic silicon. Hands-on with INT8 quantization-aware training, custom CUDA C++ kernels, and creating neuromorphic models for real-time autonomous inference at the edge. Strong foundation in computer vision, high-performance computing, and real-time embedded systems, plus production LLM/agentic systems.

TOOLS & SKILLS

Edge AI & Optimization: QAT, INT8/FP16, Model Surgery, Pruning, BrainChip Akida, NVIDIA Jetson, Neuromorphic/SNNs, ONNX Runtime

HPC & GPU: CUDA C++, cuBLAS, NVIDIA Nsight Compute, Roofline/CGMA Analysis, CMake

ML & Computer Vision: PyTorch, TensorFlow, Semantic Segmentation, RWKV/Sequence Modeling, Sensor Fusion, Imitation Learning, Depth Estimation, Reinforcement Learning

Autonomy & Robotics: CARLA, ROS2, Autonomous Vehicles, Simulink/Quanser Control (PID)

Embedded & Real-Time: C/C++, FreeRTOS, STM32, RTOS, MATLAB/Simulink

LLM Systems & Infra: FastAPI, PostgreSQL, LiteLLM, LangGraph, MCP, Docker, Cloudflare Zero Trust, Linux, GitHub Actions

Languages: Python, C/C++, SQL, MATLAB

WORK EXPERIENCE

Machine Learning Engineer Intern — **May 2025 - September 2025**
BrainChip — *Remote (CA)*

- Built an end-to-end PyTorch→Akida pipeline for high-resolution semantic segmentation on the BrainChip Akida 2.0 neuromorphic processor (8-bit integer inference).
- Performed model customization (stripping batch-norm and transposed convolutions) to map MobileNetV2/V4 backbones onto the fabric; reached ~82% of SOTA accuracy at 7.5% of the parameters (4.6M vs 61.3M).
- Integrated QAT (`quantizeml`, `cnn2snn`) to convert float32→INT8 `.fbz` models; used halo-tiling for full-resolution edge inference; trained multi-GPU (RTX A6000) on Cityscapes (~0.55 mIoU) and Portrait128 (~0.97 IoU).

Teaching Assistant - Real-Time & Embedded Systems — **August 2025 - May 2026**
Rochester Institute of Technology — *Rochester, NY*

- Guided students through core embedded concepts: RTOS, PWM, GPIO, DMA, and USART in C.

Teaching Assistant - Introduction to Generative AI — **February 2025 - May 2025**
Rochester Institute of Technology — *Rochester, NY*

- Developed the curriculum (syllabus, lectures, labs) for RIT's new Generative AI course under the instructor, covering CNNs, GANs, LLMs, and diffusion models.

Teaching Assistant - Classical Control Systems — **September 2025 - December 2025**
Rochester Institute of Technology — *Rochester, NY*

- Led MATLAB & Quanser Qube-Servo2 labs; taught feedback control, second-order systems, and PID tuning.

TECHNICAL PROJECTS

RRI Orchestrator - Production Multi-Agent LLM Platform — **October 2025 - May 2026**
Rochester Institute of Technology (RIT) — *Rochester, NY*

- Built a full-stack multi-agent platform decoupling LLM cognition from ROS2-ready robot endpoints: async FastAPI, LiteLLM routing (GPT-4o/Gemini/Claude), PostgreSQL, secured with Cloudflare Zero Trust + RBAC.
- Prototyped a cyclic LangGraph + MCP refactor (SQLite/PostgreSQL MCP servers, human-in-the-loop).

Neuromorphic Vision-RWKV - Autonomous Vehicle Control — **January 2026 - May 2026**
Rochester Institute of Technology (RIT) — *Rochester, NY*

- Built an end-to-end RWKV-6 (linear-attention) driving controller with O(1) constant-state inference - 6.88 ms median latency on RTX 3080 - targeting NVIDIA Jetson and neuromorphic edge.
- Engineered a 3.6 TB CARLA data pipeline (3.6M frames, versioned HDF5); 3-camera input → gated navigation fusion → steering/throttle heads; 100+ pytest tests, GitHub Actions CI, pre-commit (black/flake8/mypy).

VGG-16 Inference Optimization (CUDA) — **March 2025 - May 2025**
Rochester Institute of Technology (RIT) — *Rochester, NY*

- Custom CUDA C++ kernels (shared-memory tiling, memory coalescing, cuBLAS) for conv & FC layers, profiled with Nsight Compute to ~41.5 GFLOPS at batch=16.
- CGMA/roofline analysis identified FC layers as memory-bound; ONNX Runtime for model loading.

Quantum Policy Gradient for CartPole — **October 2025 - December 2025**
Rochester Institute of Technology (RIT) — *Rochester, NY*

- Built a 4-qubit variational quantum-circuit RL agent (PennyLane) vs. a parameter-matched classical MLP; statistical parity at 18% fewer parameters (paired t-tests, 9 seeds, 95% CIs).

Autonomous Driving, CARLA Sim — **September 2024 - December 2024**
Rochester Institute of Technology (RIT) — *Rochester, NY*

- Fused RGB/depth/collision sensors for real-time perception; Ultrafast lane detection, YOLOv8 object detection, traffic-light recognition, waypoint nav.

RESEARCH EXPERIENCE

Resource-Efficient Multimodal Models | *Rochester Institute of Technology, NY* — **February 2025 - May 2025**

- Model compression for edge LLM/VLM deployment: structured pruning (LLM-Pruner/LoRAPrune), quantization (GPTQ/AWQ/SpQR), efficient encoders.

NSF AWARE-AI - Graduate Research Trainee | *Rochester Institute of Technology, NY* — **April 2025 - May 2026**

- Semi-supervised 3D monocular depth estimation as low-cost LiDAR alternative; benchmarked MiDaS and Depth-Anything.

Graduate Researcher - Social Robotics | *Rochester Institute of Technology, NY* — **September 2025 - May 2026**

- HRI + user-trust calibration in LLM interactions; built the group's multi-agent orchestrator.

EDUCATION

Rochester Institute of Technology - Computer Engineering, M.S. (GPA 3.70) — **August 2024 - May 2026**
Relevant Courses: *High Performance Architectures*, *Robot Perception*, *Quantum Machine Learning*, *Real-time & Embedded Systems*, *Principles of Robotics*, & *Self-Adaptive Software w/ RL*.

Wartburg College, Iowa - Computer Science & Actuarial Science, B.A. — **May 2020**
Minors: Data Analytics & Economics